## NAME

**mbhistory** − List, compare, tag, prune and restore the versions of edit save files.

## VERSION

Version 5.0

## SYNOPSIS

**mbhistory** −−**input**=*file* {−**I***file*} [−−**list** −−**show**=*version* −−**diff**=*version*[**:***version*] −−**restore**=*version* −−**restore−tag**=*name* −−**restore−date**=*yyyy−mm−dd*[**T***hh:mm*[**:***ss*]] −−**tag**=*name* −−**prune** −−**keep**=*number* −−**older−than**=*days* −−**dry−run** −−**yes** −−**force** −−**process** −−**verbose** −−**help**]

## DESCRIPTION

In **MB-System**, bathymetry edits made with **mbedit**, **mbeditviz**, **mbclean**, **mbareaclean** and other programs are stored in binary "edit save files" (file suffix ".esf"). Before any of these programs rewrites or appends to an existing esf file, the MBIO library saves a copy of it (a snapshot) in a directory named **mbhistory** in the same directory as the esf file (normally the directory of the swath file). The program **mbhistory** lists these versions, compares them, labels them with tags, removes old ones, and restores an earlier version, so that edits can be reverted to a previous state of a single file or of a whole survey.

Each snapshot is named after the esf file, followed by a version number and the UTC date and time at which it was saved:

mbhistory/0001_20261020.mb88.esf.v0003_20261021T091530

Each esf file also has a text manifest (for example mbhistory/0001_20261020.mb88.esf.manifest) with one line per version giving the version number, date, the program that was about to modify the file, user, host, size, CRC-32 checksum, snapshot name and an optional note. Tags are stored in mbhistory/tags.txt. A snapshot holds the state of the edits at the time it was saved, just before the named program changed them. The current esf file is always the newest state and is called **now**. A program run that reopens the same esf file several times saves at most one snapshot of it, and no snapshot is saved if the file is unchanged since the newest one. Every rewrite of an esf file updates the program, user, cpu and date lines of its header, so a rewrite that leaves the edits and the ESF mode unchanged also counts as unchanged.

When a program creates a new esf file, the empty file (no edits) is saved as the first version with the note "created", so the history also shows when editing of the swath file began. Esf files created before this was done have no such version; their history starts with the first rewrite.

Restoring never deletes versions. Before a version is restored, the current esf file is saved as a new version with the note "restore (vN −> vM)", where vN is that new version and vM the restored one, so a restore can itself be undone. Only −−**prune** removes versions, and it keeps tagged versions unless −−**force** is given. After a restore the processed swath file is out of date: run **mbprocess**, or use −−**process**.

The input may be a single swath file or a datalist. With a datalist each command is applied to every swath file in it; for example −−**restore−date** returns every file to the state it had at that time, and files that were not changed since are left alone.

Snapshots are not saved if the environment variable **MB_HISTORY** is set to "off" or "0".

## MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping

University of New Hampshire
Christian dos Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

## MBHISTORY OPTIONS

**−−input**=*file* **−I***file*
Swath file or datalist whose edit save file versions are handled. The esf file of a swath file is the one named in its mbprocess parameter file, or otherwise the swath file name followed by ".esf".

**−−list** Lists the versions with their date, the program that was about to modify the file, the user, the number of edit events, tags and notes, followed by the current file. This is the default.

**−−show**=*version*
Shows the details of one version (a number, or **now** for the current file), including the counts of each edit action and the header stored in the esf file.

**−−diff**=*version*[**:***version*]
Compares the final edit of every beam between two versions (the second one defaults to **now**) and reports how many beams are now flagged, now unflagged, have a different flag type or no longer have an edit, and the pings affected. With **−−verbose** every changed beam is listed.

**−−restore**=*version*
Restores the given version as the current esf file.

**−−restore−tag**=*name*
Restores the version that carries the tag *name*.

**−−restore−date**=*yyyy−mm−dd*[**T***hh:mm*[**:***ss*]]
Restores the edits as they were at the given UTC date and time, which is the first version saved at or after that time; if no version was saved since, the current file is already in that state and nothing is changed. A date without a time means the end of that day. A date before the "created" version restores an esf file with no edits. If the oldest version of a file is not a "created" version and the date is before it, the edits made before that version were not recorded: the oldest version is restored and a warning is printed.

**−−tag**=*name*
Labels the current state with the one-word tag *name*. A snapshot is saved if the current file differs from the newest version. An existing tag of the same name is moved only with **−−force**.

**−−prune**
Removes old versions selected with **−−keep** and/or **−−older−than** (a version is removed only if all given conditions select it). Tagged versions are kept unless **−−force** is given.

**−−keep**=*number*
With **−−prune**, keeps the newest *number* versions.

**−−older−than**=*days*
With **−−prune**, removes only versions older than *days* days.

**−−dry−run**
Shows what a restore or prune would do without changing anything.

**−−yes** Restores and prunes change files only when **−−yes** is given; without it they run as a dry run.

**−−force**
Allows **−−prune** to remove tagged versions and **−−tag** to move an existing tag.

**−−process**
Runs **mbprocess −−force** on each restored swath file, so that the processed file is regenerated even if its time stamp looks up to date.

**−−verbose −V**
Increases the amount of output; with **−−diff** and restores, every changed beam is listed.

**−−help −H**
This "help" flag causes the program to print out a description of its operation and then exit immediately.

## EXAMPLES

To list the versions of the edits of one swath file:

```
mbhistory −I 0001_20261020.mb88
```

To see what changed between version 2 and the current edits:

```
mbhistory −I 0001_20261020.mb88 −−diff=2
```

To return every file of a survey to the state of 21 October 2026 at 12:00 UTC, first as a dry run and then for real, reprocessing the files:

```
mbhistory −I datalist.mb−1 −−restore−date=2026−10−21T12:00
mbhistory −I datalist.mb−1 −−restore−date=2026−10−21T12:00 −−yes −−process
```

To tag the current edits of a survey before a risky operation, and to return to them later:

```
mbhistory −I datalist.mb−1 −−tag=before−areaclean
mbhistory −I datalist.mb−1 −−restore−tag=before−areaclean −−yes
```

To keep only the 20 newest versions of each file:

```
mbhistory −I datalist.mb−1 −−prune −−keep=20 −−yes
```

## SEE ALSO

**mbsystem**(1), **mbedit**(1), **mbeditviz**(1), **mbclean**(1), **mbareaclean**(1), **mbprocess**(1), **mbdumpesf**(1)

## BUGS

Versions are matched to swath files by file name, so renaming a swath file detaches it from its history.